

FUNCTIONS OF PLACENTA

Headings

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1. NUTRITIVE FUNCTION

2. EXCRETORY FUNCTION

▼ 3. RESPIRATORY FUNCTION

Exchange of Respiratory Gases

Bohr Effect & Double Bohr Effect

▼ 4. ENDOCRINE FUNCTION

Detailed Hormonal Actions

FETOPLACENTAL UNIT

- **Placenta** : A temporary membranous vascular organ that develops in females during pregnancy.
- **Key Functions :—**
 - a. Nutritive function
 - b. Excretory function
 - c. Respiratory function
 - d. Endocrine function

1. NUTRITIVE FUNCTION

Nutritive substances, electrolytes, and hormones necessary for fetal development



Diffuse directly from mother's blood across the placental barrier into fetal blood

2. EXCRETORY FUNCTION

Metabolic end products and other waste products from the fetal body



Excreted into maternal blood for elimination via maternal kidneys

3. RESPIRATORY FUNCTION

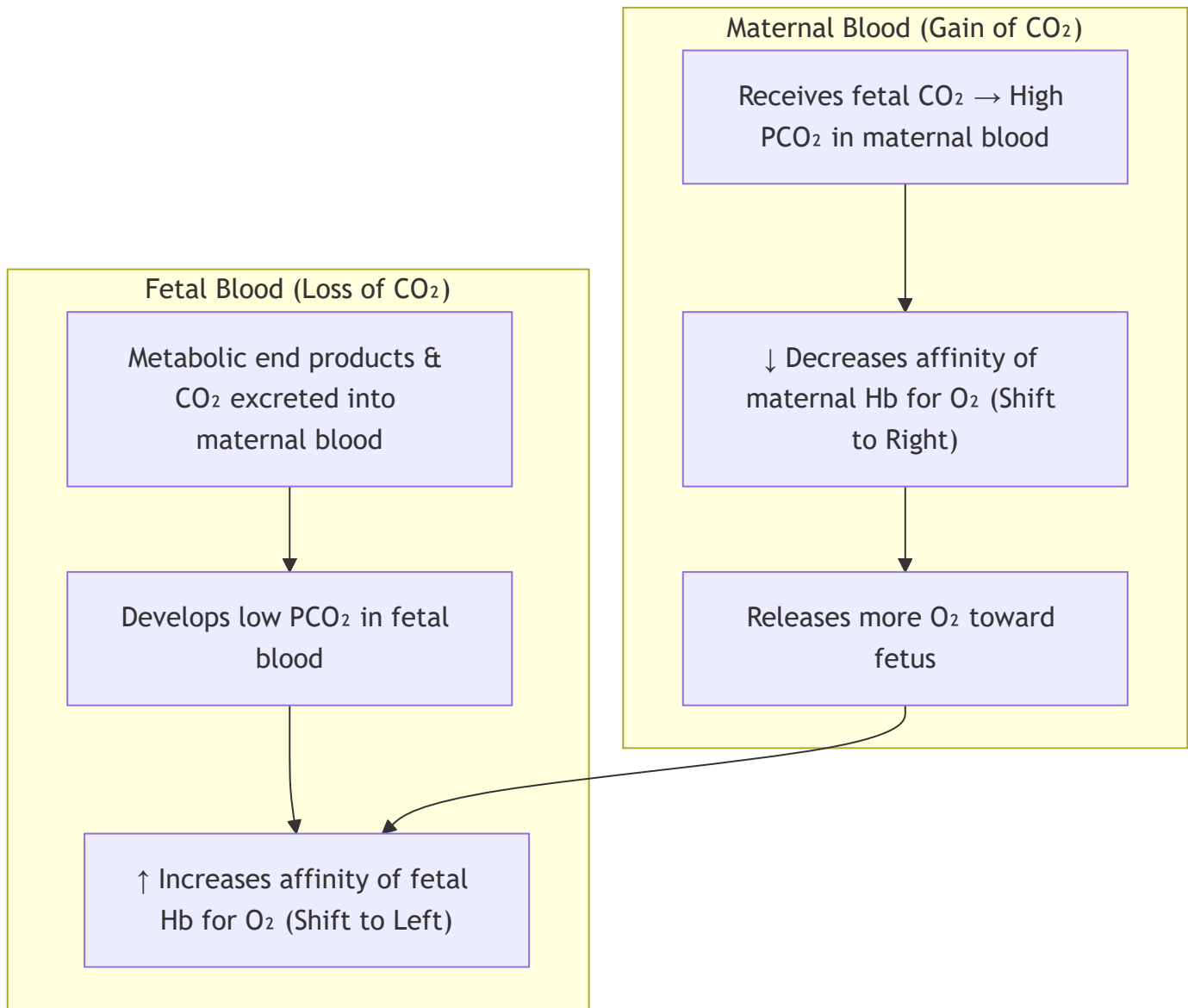
- Fetal lungs are **non-functioning**; the placenta serves as the respiratory organ for the fetus.
- Necessary O_2 is received by diffusion from maternal blood.

Exchange of Respiratory Gases

- Occurs via a **partial pressure gradient**:
 - PO_2 in maternal blood = 50 mmHg
 - PO_2 in fetal blood = 30 mmHg
 - **Pressure gradient of 20 mmHg** drives the diffusion of O_2 into fetal circulation.
- **Adequate Oxygen Delivery Factors :—**
 - a. **Fetal Hemoglobin (HbF)**: Has 20 times higher affinity for oxygen than adult hemoglobin (HbA).
 - b. **Hemoglobin Concentration**: Approximately 50% higher in fetal blood compared to adult blood.

Bohr Effect & Double Bohr Effect

Bohr Effect: ↓ Decrease in Hb affinity for O_2 due to ↑ increased CO_2 tension (acidosis)



Double Bohr Effect : The simultaneous operation of the Bohr effect in both fetal blood (left shift) and maternal blood (right shift), maximizing oxygen transfer across the placenta.

4. ENDOCRINE FUNCTION

The placenta secretes **5 major hormones**:

- i. **Human Chorionic Gonadotropin (hCG)**
- ii. **Estrogen**
- iii. **Progesterone**
- iv. **Human Chorionic Somatomammotropin (HCS / Placental Lactogen)**
- v. **Relaxin**

Detailed Hormonal Actions

FETOPLACENTAL UNIT

- **Definition :** Functional cooperation and metabolic interaction between the **fetus** and the **placenta** in the synthesis of steroid hormones.
- **Reason for Interaction :** Some key steroidogenic enzymes are present only in the fetus and absent in the placenta, and vice versa.

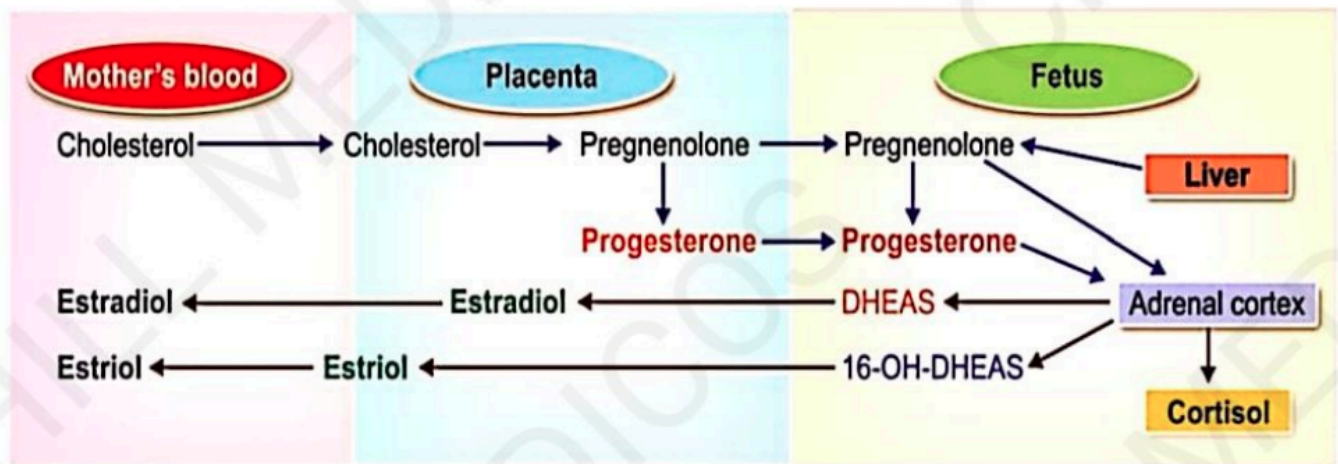


FIGURE 85.1: Fetoplacental unit. DHEAS = Dehydroepiandrosterone sulfate, 16-OH-DHEAS = 16-hydroxy-dehydroepiandrosterone sulfate.